



**Stan's
Technologies**

Benchmark Report

BoolMinGeo vs PyEDA Boolean Minimization

Experiment Date: 2026-01-30

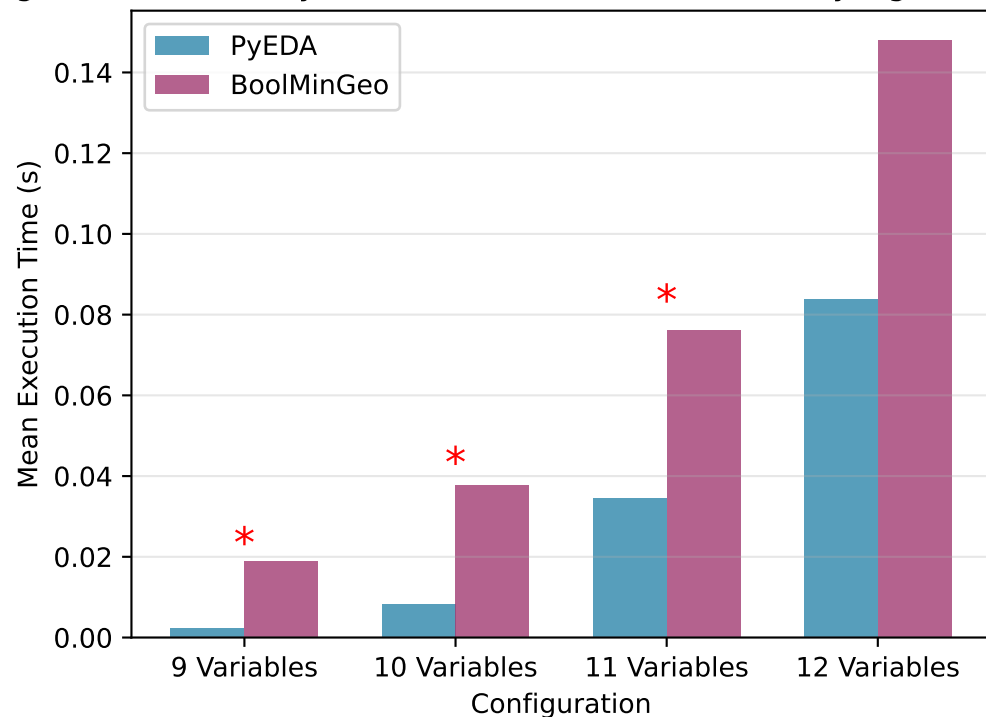
Random Seed: 42

Total Test Cases: 12

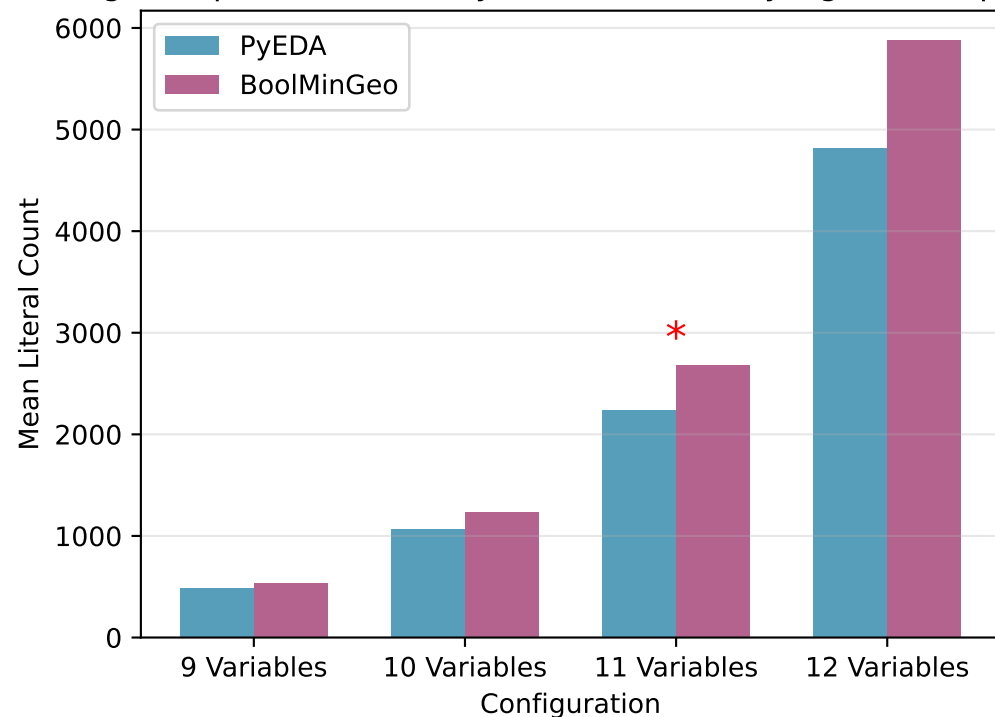
Statistical Significance Level: $\alpha = 0.05$

A Statistical Comparison of Boolean Minimization Algorithms

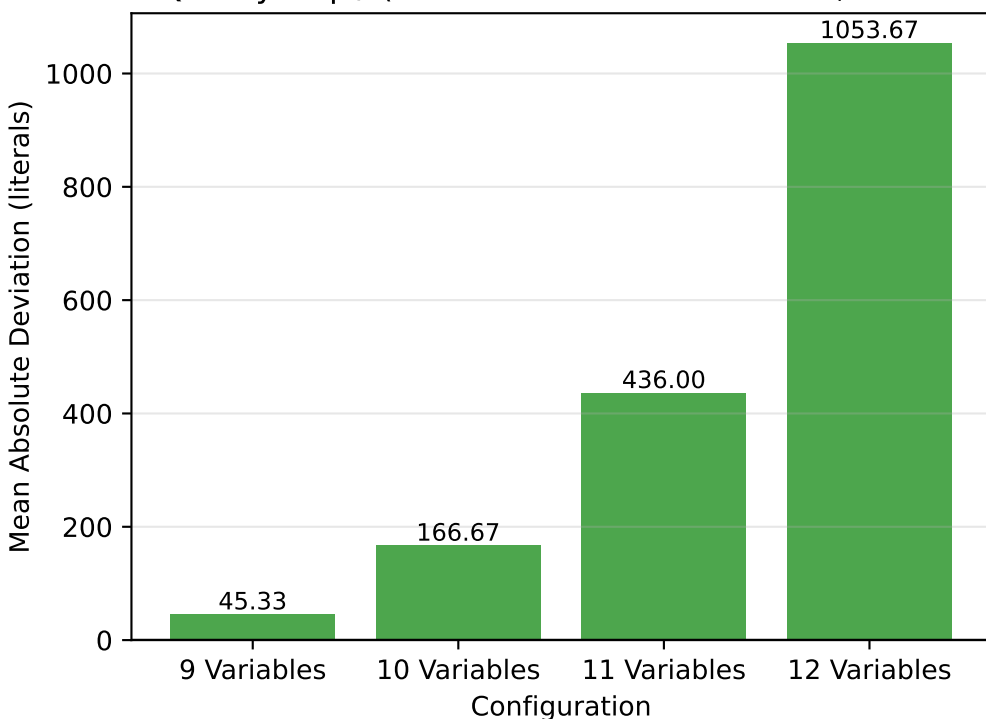
Average Performance by Variable Count\n(* = statistically significant, p < 0.05)



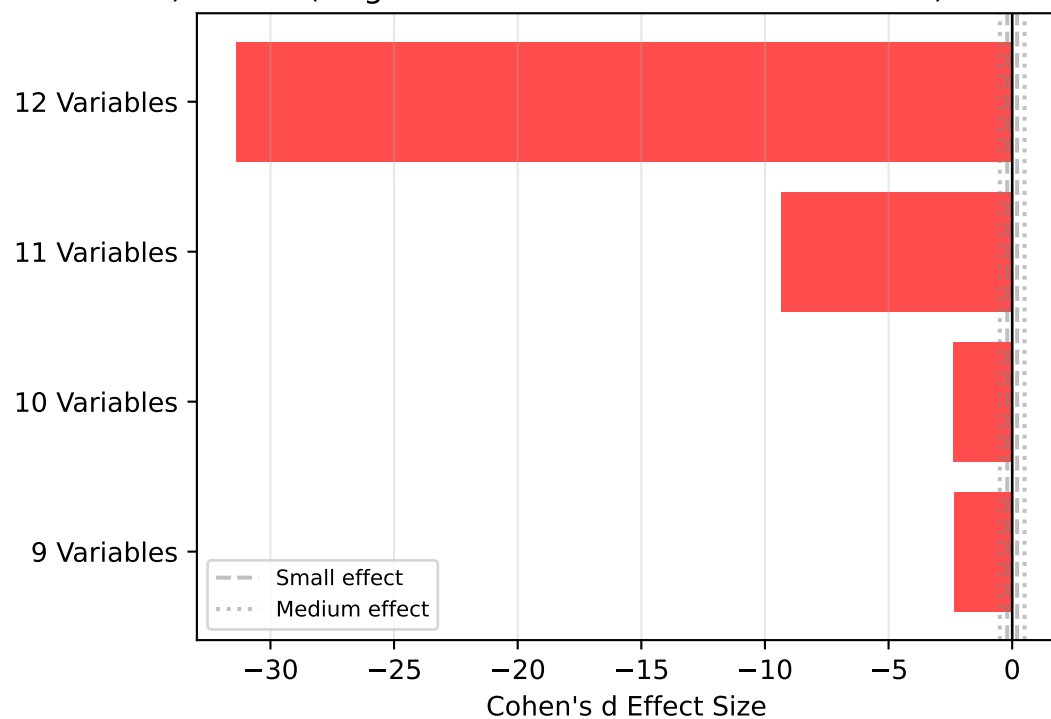
Average Simplification Quality\n(* = statistically significant, p < 0.05)



Simplification Quality Gap\n(Green = BoolMinGeo better, Red = PyEDA better)



Effect Size: Simplification Quality
(Negative = BoolMinGeo more minimal)



STATISTICAL ANALYSIS SUMMARY

EXPERIMENT OVERVIEW

=====
Total Test Cases: 12
Configurations Tested: 4 (9-10 variables)
Tests per Config: 10 (sparse, balanced, dense distributions)
Random Seed: 42
Significance Level: $\alpha = 0.05$

9-VARIABLE FUNCTIONS (3 tests, 3 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.002405 s
Mean BoolMinGeo: 0.018882 s
Mean Difference: -0.016478 s
95% CI: [-0.018503, -0.014452]
p-value: 0.000815
Statistically Sig: YES
Effect Size (d): -20.2092 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 489.00 literals
Mean BoolMinGeo: 534.33 literals
Mean Difference: -45.33 literals
Mean Deviation: 45.33 literals
Quality Gap: 9.27% relative to PyEDA
95% CI: [-93.07, 2.41]
p-value: 0.055009
Statistically Sig: NO
Effect Size (d): -2.3589 (large)
Winner: PyEDA

10-VARIABLE FUNCTIONS (3 tests, 3 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.008347 s
Mean BoolMinGeo: 0.037824 s
Mean Difference: -0.029477 s
95% CI: [-0.037521, -0.021433]
p-value: 0.003999
Statistically Sig: YES
Effect Size (d): -9.1029 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 1072.33 literals
Mean BoolMinGeo: 1239.00 literals
Mean Difference: -166.67 literals
Mean Deviation: 166.67 literals
Quality Gap: 15.54% relative to PyEDA
95% CI: [-340.95, 7.62]
p-value: 0.054302
Statistically Sig: NO
Effect Size (d): -2.3755 (large)
Winner: PyEDA

11-VARIABLE FUNCTIONS (3 tests, 3 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.034496 s
Mean BoolMinGeo: 0.076131 s
Mean Difference: -0.041635 s
95% CI: [-0.072839, -0.010431]
p-value: 0.029027
Statistically Sig: YES
Effect Size (d): -3.3145 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 2241.67 literals
Mean BoolMinGeo: 2677.67 literals
Mean Difference: -436.00 literals
Mean Deviation: 436.00 literals
Quality Gap: 19.45% relative to PyEDA
95% CI: [-552.01, -319.99]
p-value: 0.003803
Statistically Sig: YES
Effect Size (d): -9.3360 (large)
Winner: PyEDA

12-VARIABLE FUNCTIONS (3 tests, 3 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.083934 s
Mean BoolMinGeo: 0.147909 s
Mean Difference: -0.063975 s
95% CI: [-0.075613, -0.052337]
p-value: 0.001783
Statistically Sig: YES
Effect Size (d): -13.6552 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 4822.67 literals
Mean BoolMinGeo: 5876.33 literals
Mean Difference: -1053.67 literals
Mean Deviation: 1053.67 literals
Quality Gap: 21.85% relative to PyEDA
95% CI: [-1137.04, -970.30]
p-value: 0.000338
Statistically Sig: YES
Effect Size (d): -31.3957 (large)
Winner: PyEDA

CONCLUSIONS

KEY FINDINGS

- 1. PERFORMANCE:
 - Average PyEDA time: 0.032295 s
 - Average BoolMinGeo time: 0.070187 s
 - Difference: -0.037891 s
 - Statistical significance: YES (p = 0.000028)
- 2. SIMPLIFICATION QUALITY:
 - Average PyEDA literals: 2156.42
 - Average BoolMinGeo literals: 2581.83
 - Mean deviation: 425.42 literals
 - Quality gap: 19.73% relative to PyEDA
 - Statistical significance: YES (p = 0.004117)
- 3. EQUIVALENCE:
 - All 12/12 tests passed logical equivalence checks
 - Both algorithms produce functionally correct results
- 4. EFFECT SIZES:
 - Performance: -1.9740 (large)
 - Simplification: -1.0413 (large)

INTERPRETATION

Per-Variable Analysis:

- 9 variables: 9.27% quality gap, PyEDA produces more minimal results (Mean deviation: 45.33 literals)
- 10 variables: 15.54% quality gap, PyEDA produces more minimal results (Mean deviation: 166.67 literals)
- 11 variables: 19.45% quality gap, PyEDA produces more minimal results (Mean deviation: 436.00 literals)
- 12 variables: 21.85% quality gap, PyEDA produces more minimal results (Mean deviation: 1053.67 literals)

REPRODUCIBILITY

- Random seed: 42
- All test cases can be reproduced with this seed
- Statistical tests use standard methods (paired t-test, Cohen's d)

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